

Chapter 1, Section 3

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2. A morphism whose underlying map on the topological space is a homeomorphism need not be an isomorphism.

- (a) For example, let $\varphi : \mathbb{A}^1 \rightarrow \mathbb{A}^2$ be defined by $t \mapsto (t^2, t^3)$. Show that φ defines a bijective bicontinuous morphism of $\mathbb{A}^1$ onto the curve $y^2 = x^3$, but that φ is not an isomorphism.
- (b) For another example, let the characteristic of the base field k be $p > 0$, and define a map $\varphi : \mathbb{A}^1 \rightarrow \mathbb{A}^1$ by $t \mapsto t^p$. Show that φ is bijective and bicontinuous but not an isomorphism. This is called the *Frobenius morphism*.

Proof.

- (a) Let Y be the curve $y^2 = x^3$. Then φ clearly maps into Y and is injective since

$$(t^2, t^3) = (u^2, u^3) \implies tu^2 = t^2u \implies t = u.$$

If $P = (a_1, a_2) \in Y$, then clearly $\varphi(\sqrt[3]{a_2}) = P$ since k is algebraically closed, hence φ is bijective onto Y . To show φ is bicontinuous it is enough to show φ and φ^{-1} of closed sets in $\mathbb{A}^1$ and Y , respectively, is closed. A closed set in either $\mathbb{A}^1$ or Y is a finite set of points, so by the bijectivity of φ , it is also bicontinuous.

To show it is not an isomorphism, it suffices to show the coordinate rings of the affine line $\mathbb{A}^1$ and the curve Y are not isomorphic. Indeed, $A(\mathbb{A}^1) = k[x]$ and $A(Y) = k[x, y]/(y^2 - x^3)$, and while $k[x]$ is factorial, $A(Y)$ is not since $y^2 = x^3$.

- (b) It is bijective and bicontinuous for the same reason in (a). If φ is an isomorphism, then it must induce an automorphism $\varphi^* : k[x] \rightarrow k[x]$ of the affine coordinate ring $k[x]$ of $\mathbb{A}^1$. However, φ^* is defined by $x \mapsto x^p$, which is not even surjective.

□

- 3. (a) Let $\varphi : X \rightarrow Y$ be a morphism. Then for each $P \in X$, φ induces a homomorphism of local rings $\varphi_P^* : \mathcal{O}_{\varphi(P), Y} \rightarrow \mathcal{O}_{P, X}$.
- (b) Show that a morphism φ is an isomorphism if and only if φ is a homeomorphism, and that the induced map φ_P^* on local rings is an isomorphism for all $P \in X$.
- (c) Show that if $\varphi(X)$ is dense in Y , then the map φ_P^* is *injective* for all $P \in X$.

Proof.

- (a) We have already seen that φ induces a homomorphism $\varphi^* : \mathcal{O}(Y) \rightarrow \mathcal{O}(X)$. By (2.3), there exists an open affine variety V in Y containing $\varphi(P)$. Then $U = \varphi^{-1}(V)$ contains P , so $\mathcal{O}_{P, X} \simeq \mathcal{O}_{P, U}$ and $\mathcal{O}_{\varphi(P), Y} \simeq \mathcal{O}_{\varphi(P), V}$. By definition f is a regular function defined in some open neighborhood of $\varphi(P)$ in V such that $f(\varphi(P))$ is nonzero if and only if $\varphi^* f$ is nonzero at P . Thus, $\varphi^* f$ is a unit in $\mathcal{O}_{P, U}$, so by the universal property of localization there exists a unique ring homeomorphism $\varphi_P^* : \mathcal{O}_{\varphi(P), Y} \rightarrow \mathcal{O}_{P, X}$.
- (b) The only if direction is obvious. Conversely, if φ_P^* on local rings is an isomorphism for all $P \in X$, then since isomorphism is a local property (A.M. p. 40), we have an isomorphism $\varphi^* : \mathcal{O}_V \rightarrow \mathcal{O}_U$, where V is any affine variety in Y and $U = \varphi^{-1}(V)$. Hence, $\varphi : U \rightarrow V$ is an isomorphism.
- (c) We want to show for any $P \in X$ that if f, g are elements of $\mathcal{O}_{\varphi(P), Y}$ such that $\varphi_P^* f = \varphi_P^* g$ on some open neighborhood U of P , then $f = g$ in some open neighborhood V of $\varphi(P)$. Let V be an open neighborhood of $\varphi(P)$ with both f and g defined so that $\varphi_P^* f$ and $\varphi_P^* g$ are defined on $U = \varphi^{-1}(V)$. Then, $\varphi(U) = V \cap \varphi(X)$ is a closed and dense subset of V such that $f = g$, hence $f = g$ on V .

□

4. Show that the d -uple embedding of $\mathbb{P}^n$ (I, Ex. 2.12) is an isomorphism onto its image.

Proof. Let $\rho : \mathbb{P}^n \rightarrow Y \subseteq \mathbb{P}^N$ be the d -uple embedding of $\mathbb{P}^n$ with $N = \binom{n+d}{d} - 1$ and image restricted so that ρ is a homeomorphism. By Exercise 2, to show ρ is an isomorphism it suffices to show it is an isomorphism locally. Let $P = (a_0, \dots, a_n) \in \mathbb{P}^n$ and $\rho(P) = (b_0, \dots, b_N) \in Y$ and assume $a_0 \neq 0$. By (3.4), we have the following identifications of local rings

$$\mathcal{O}_{P, \mathbb{P}^n} = S(\mathbb{P}^n)_{(\mathfrak{m}_P)}, \quad \mathcal{O}_{\rho(P), Y} = S(Y)_{(\mathfrak{m}_{\rho(P)})},$$

where $\mathfrak{m}_P$ (respectively, $\mathfrak{m}_{\rho(P)}$) is the ideal generated by the set of homogenous $f \in S(\mathbb{P}^n)$ (respectively, $f \in S(Y)$) such that $f(P) = 0$ (respectively, $f(\rho(P)) = 0$). We want to show the induced map $\rho_P^* : \mathcal{O}_{\rho(P), Y} \rightarrow \mathcal{O}_{P, \mathbb{P}^n}$ is an isomorphism. By the identification above, the map ρ_P^* can be defined by the canonical projection map $\bar{\theta} : k[y_0, \dots, y_N]/\mathfrak{a} \rightarrow k[x_0, \dots, x_n]$ induced by θ in Exercise 2.12 as

$$\rho_P^*\left(\frac{f}{g}\right) = \frac{\bar{\theta}(f)}{\bar{\theta}(g)},$$

where f and g are homogenous elements of same degree in $S(Y)$ such that $g(\rho(P)) \neq 0$ (since we are only concerned with homogenous elements of degree 0 in $S(Y)_{\mathfrak{m}_{\rho(P)}}$). Clearly ρ_P^* is injective since $\bar{\theta}$ is injective, and ρ_P^* maps $S(Y)_{(\mathfrak{m}_{\rho(P)})}$ into $S(\mathbb{P}^n)_{(\mathfrak{m}_P)}$ since $\theta(y_i)$ is of the same degree for all $0 \leq i \leq N$ and $\mathfrak{a}$ is a homogenous ideal. To show it is surjective, suppose $h/k \in \mathcal{O}_{P, \mathbb{P}^n}$ with $k(P) \neq 0$ and h, k both homogenous of degree e . We can assume e is a multiple d since x_0 is a unit in $\mathcal{O}_{P, \mathbb{P}^n}$, and we can multiply h/k by a sufficient power of x_0/x_0 to obtain the desired degrees in the numerator and denominator. Then, each term in h and k is the product of monomials of degree d , and since $\bar{\theta}$ is surjective onto such elements, ρ_P^* is surjective. Hence, ρ_P^* is an isomorphism. □

6. There are quasi-affine varieties which are not affine. For example, show that $X = \mathbb{A}^2 - \{(0, 0)\}$ is not affine.

Proof. The ideal $I(X)$ consists of all polynomials in $k[x, y]$ that vanish at X . If $f \in I(X)$, then viewing f as a regular function on $\mathbb{A}^2$, $f^{-1}(0)$ is a closed and dense subset of $\mathbb{A}^2$, hence it equals to $\mathbb{A}^2$, hence $I(X) = (0)$. Thus, if X is affine, then

$$\mathcal{O}(X) = k[x, y]/I(X) \simeq k[x, y] = \mathcal{O}(\mathbb{A}^2),$$

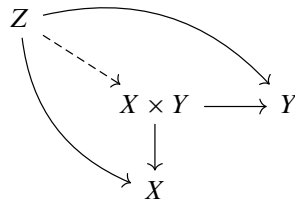
so by (3.5) and (3.7) $X \cong \mathbb{A}^2$ under the inclusion map, which is ridiculous. □

12. If P is a point on a variety X , then $\dim \mathcal{O}_P = \dim X$.

Proof. We can find an open affine variety Y in X containing P , and since local rings behave well under open subsets, i.e. $\mathcal{O}_{P, Y} \simeq \mathcal{O}_{P, X}$, we can reduce to the affine case. Then, the statement follows from (3.2). □

15. *Products of Affine Varieties.* Let $X \subseteq \mathbb{A}^n$ and $Y \subseteq \mathbb{A}^m$ be affine varieties.

- Show that $X \times Y \subseteq \mathbb{A}^{n+m}$ with its induced topology is irreducible. The affine variety $X \times Y$ is called the *product* of X and Y . Note that its topology is in general not equal to the product topology.
- Show that $A(X \times Y) \simeq A(X) \otimes_k A(Y)$.
- Show that $X \times Y$ is a product in the category of varieties.
- Show that $\dim X \times Y = \dim X + \dim Y$.



Proof.

- (a) Suppose that $X \times Y$ is a union of two closed subsets $Z_1 \cup Z_2$. Let $X_i = \{x \in X \mid x \times Y \subseteq Z_i\}$, $i = 1, 2$ and let $\pi : \mathbb{A}^{n+m} \rightarrow \mathbb{A}^m$ be the projection map. For any $x \in X$ since $x \times Y \subseteq Z_1 \cup Z_2$, we can write $x \times Y = (x \times Y \cap Z_1) \cup (x \times Y \cap Z_2)$, so let $C_i = x \times Y \cap Z_i$, then

$$Y = \pi(x \times Y) = \pi(C_1 \cup C_2) = \pi(C_1) \cup \pi(C_2).$$

We can write $Z_i = Z(f_1, \dots, f_r)$, $i = 1, 2$, for some $f_j \in A(\mathbb{A}^{n+m})$, then we see that

$$\pi(C_i) = \bigcap_{j=1}^r Z(f_j(x, y_1, \dots, y_m)),$$

where we can view $f_j(x, y_1, \dots, y_m)$ as an element in $A(\mathbb{A}^m)$, which shows $\pi_2(C_i)$ is a closed set in $\mathbb{A}^m$. Since Y is irreducible, we have $\pi(C_2) = \emptyset$, say, hence $Y = \pi(C_1)$, hence $x \times Y = x \times Y \cap Z_1$, hence $x \times Y \subseteq Z_1$. In particular, we have $X = X_1 \cup X_2$, and we have shown $Z_i = X_i \times Y$. We can repeat the argument for $Y_i = \{y \in Y \mid X \times Y \subseteq Z_i\}$, $i = 1, 2$ so that $Z_i = X \times Y_i$. This is only possible if $X = X_1$ and $Y = Y_1$, say, hence $Z_1 = X \times Y$.

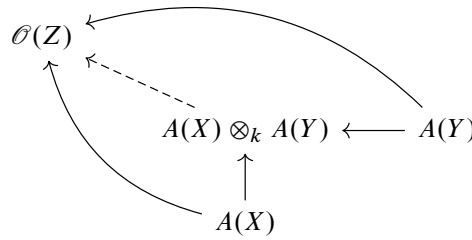
- (b) We clearly have $A(\mathbb{A}^{n+m}) \simeq A(\mathbb{A}^n) \otimes_k A(\mathbb{A}^m)$, where $A(\mathbb{A}^n) = k[x_1, \dots, x_n]$ and $A(\mathbb{A}^m) = k[y_1, \dots, y_m]$. Then, we have

$$I(X \times Y) = I(X) \otimes_k A(\mathbb{A}^m) + A(\mathbb{A}^n) \otimes_k I(Y),$$

hence we have the isomorphism

$$\begin{aligned} A(X \times Y) &\cong \frac{A(\mathbb{A}^n) \otimes_k A(\mathbb{A}^m)}{I(X \times Y)} \\ &\cong \frac{A(\mathbb{A}^n) \otimes_k A(\mathbb{A}^m)}{I(X) \otimes_k A(\mathbb{A}^m) + A(\mathbb{A}^n) \otimes_k I(Y)} \\ &\cong \frac{A(\mathbb{A}^n)}{I(X)} \otimes_k \frac{A(\mathbb{A}^m)}{I(Y)} \\ &\cong A(X) \otimes_k A(Y). \end{aligned}$$

- (c) Suppose we have morphisms $Z \rightarrow X$ and $Z \rightarrow Y$. This induces k -algebra homomorphisms $A(X) \rightarrow \mathcal{O}(Z)$ and $A(Y) \rightarrow \mathcal{O}(Z)$, which defines a k -bilinear map $A(X) \times A(Y) \rightarrow \mathcal{O}(Z)$. By the universal property of tensor products, we have a unique k -algebra homomorphism $A(X) \otimes_k A(Y) \rightarrow \mathcal{O}(Z)$, hence we have a unique morphism $W \rightarrow X \times Y$.



- (d) If $t_1, \dots, t_n$ and $u_1, \dots, u_m$ are coordinates for X and Y , respectively, then each t_i, u_j are algebraically independent in $A(X \times Y)$, that is the quotient field of $A(X \times Y)$ has transcendence degree $n + m$, hence $\dim X \times Y = \dim X + \dim Y$.

□

17. Normal Varieties. A variety Y is *normal at a point* $P \in Y$ if $\mathcal{O}_P$ is integrally closed ring. Y is *normal* if it is normal at every point.

- Show that every conic in $\mathbb{P}^2$ is normal.
- Show that the quadric surfaces Q_1, Q_2 in $\mathbb{P}^3$ given by equations $Q_1 : xy = zw$; $Q_2 : xy = z^2$ are normal.
- Show that the cuspidal cubic $y^2 = x^3$ in $\mathbb{A}^2$ is not normal.
- If Y is affine, then Y is normal $\iff A(Y)$ is integrally closed.
- Let Y be an affine variety. Show that there is a normal affine variety $\hat{Y}$, and a morphism $\pi : \hat{Y} \rightarrow Y$, with the property that whenever Z is normal variety, and $\varphi : Z \rightarrow Y$ is a *dominant* morphism, then there is a unique morphism $\theta : Z \rightarrow \hat{Y}$ such that $\varphi = \pi \circ \theta$. $\hat{Y}$ is called the *normalization* of Y .

Proof.

- (a) A conic in $\mathbb{P}^2$ can be covered by open affine varieties isomorphic to a conic in $\mathbb{A}^2$. Thus, it suffices to show all irreducible plane algebraic curves of degree 2 are normal, that is we want to show an affine variety X in $\mathbb{A}^2$ defined by the zero set of an irreducible quadratic polynomial f in $k[x, y]$ is normal. Since being integrally closed is a local property, it suffices to show the affine coordinate ring $A(X) = k[x, y]/(f)$ is integrally closed. We first prove (Ex. 1.1c), which states $A(X)$ is isomorphic to either $R = k[x, y]/(y - x^2)$ or $S = k[x, y]/(xy - 1) \simeq k[x, \frac{1}{x}]$. For simplicity assume $\text{char } k \neq 2$. Any quadratic polynomial can be written as

$$f(x, y) = Ax^2 + Bxy + Cy^2 + Dx + Ey + F$$

We claim the following: if $B^2 - 4AC = 0$, then $A(X) \simeq R$, otherwise $A(X) \simeq S$. We proceed by showing that X can be transformed to a variety of the form $y = x^2$ or $xy = 1$ using some affine transformations.

Suppose $B^2 - 4AC = 0$, then assume $C \neq 0$ (if $C = 0$, then $B = 0$, so we already have an equation of the desired form up to some translation and stretching), so we have

$$f(x, y) = (\sqrt{A}x + \sqrt{C}y)^2 + Dx + Ey + F,$$

since k is algebraically closed. We can apply the following affine transformation

$$\begin{aligned} x &\mapsto -\sqrt{C}y \\ y &\mapsto -\sqrt{C}x + \sqrt{A}y \end{aligned}$$

which is indeed an affine transformation since it has determinant $C \neq 0$, to obtain

$$f'(x, y) = C^2x^2 - D\sqrt{C}y - E(-\sqrt{C}x + \sqrt{A}y) + F,$$

which is the case when $B = C = 0$.

Now suppose $\Delta^2 = B^2 - 4AC \neq 0$ and assume $A \neq 0$, then we have

$$f(x, y) = \left(\sqrt{A}x + \frac{B + \Delta}{2\sqrt{A}}y \right) \left(\sqrt{A}x + \frac{B - \Delta}{2\sqrt{A}}y \right) + Dx + Ey + F,$$

so we can apply the affine transformation

$$\begin{aligned} \sqrt{A}x + \frac{B + \Delta}{2\sqrt{A}}y &\mapsto x \\ \sqrt{A}x + \frac{B - \Delta}{2\sqrt{A}}y &\mapsto y \end{aligned}$$

which has determinant $1/\Delta \neq 0$, to obtain

$$f'(x, y) = xy + (\text{linear terms}).$$

Since affine coordinate rings are invariant up to affine transformations, we have either $A(X) \simeq R$ or $A(X) \simeq S$. The ring $R = k[x, y]/(y - x^2)$ is isomorphic to a polynomial ring over one variable, which is a factorial, hence it is integrally closed. The ring $S = k[x, x^{-1}]$ is a discrete valuation ring, which are integrally closed by (A.M. p. 94).

- (b) The quadric surface Q_1 can be covered by open affine varieties isomorphic to the affine surface $z = xy$. The affine coordinate ring of such surface is isomorphic to $k[x, y]$, which is integrally closed, hence Q_1 is normal. Similarly, Q_2 can be covered by open affine varieties isomorphic to either $y = x^2$ or $xy = 1$, which was shown to be normal in (a).
- (c) The affine coordinate ring $A(Y)$ is isomorphic to $k[t^2, t^3]$. The quotient field is $k(t)$, and the element $t \in k(t)$ is integral over $k[t^2, t^3]$ since $(t)^2 - t^2 = 0$ but $t \notin k[t^2, t^3]$, hence $A(Y)$ is not integrally closed, hence Y is not normal by (d).
- (d) Being integrally closed is a local property, that is a ring A is integrally closed if and only if $A_{\mathfrak{m}}$ is integrally closed for all maximal ideals $\mathfrak{m}$ in A by (A.M. p. 63). There is a one-to-one correspondence between maximal ideals of $\mathcal{O}(Y) = A(Y)$ and points of Y , hence Y is normal if and only if $\mathcal{O}_P$ is integrally closed for all $P \in Y$ if and only if $A(Y)_{\mathfrak{m}}$ is integrally closed for all maximal ideals $\mathfrak{m}$ if and only if $A(Y)$ is integrally closed.

- (e) We want to show there exists a normal scheme $\widehat{Y}$ with morphism $\pi : \widehat{Y} \rightarrow Y$ that is universal amongst all dominant morphisms from normal schemes into Y . Let $A(Y)$ be the affine coordinate ring of Y and let B be the integral closure of $A(Y)$ in its quotient field. Then, B is a finitely generated k -algebra by (3.9A), and it is a subring of a field, so it is an integral domain, which means it defines an affine variety. Let $\widehat{Y}$ be an affine variety with coordinate ring $A(\widehat{Y}) = B$, and let $\pi : \widehat{Y} \rightarrow Y$ be the morphism induced by the inclusion map $A(Y) \hookrightarrow B$. Let Z be a normal variety, and $\varphi : Z \rightarrow Y$ a dominant morphism. Since Z can be covered by open affine varieties and local rings are preserved under open subsets, we reduce to the case when Z is an affine normal variety. We want to show there exists a ring homomorphism $\alpha : \mathcal{O}(\widehat{Y}) \rightarrow \mathcal{O}(Z)$ such that the following diagram commutes:

$$\begin{array}{ccc} & \mathcal{O}(\widehat{Y}) & \\ \pi^* \nearrow & & \searrow \exists! \alpha \\ \mathcal{O}(Y) & \xrightarrow{\varphi^*} & \mathcal{O}(Z) \end{array}$$

If $\varphi^* f = 0$ for some $f \in \mathcal{O}(Y)$, then $f = 0$ on a closed dense subset of Y , hence $f = 0$ on Y , so φ^* is injective. Thus, φ^* induces a map between the function fields $\overline{\varphi^*} : K(Y) \rightarrow K(Z)$, where we have the inclusions $\mathcal{O}(Y) \hookrightarrow \mathcal{O}(\widehat{Y}) \hookrightarrow K(Y)$ by construction. If $f \in \mathcal{O}(\widehat{Y})$, then there exists an equation of integral dependence of the form

$$f^n + a_1 f^{n-1} + \cdots + a_n = 0, \quad a_i \in \mathcal{O}(Y),$$

and since $\overline{\varphi^*}(a_i) = \varphi^*(a_i) \in \mathcal{O}(Z)$, we can apply $\overline{\varphi^*}$ to the equation above to obtain an equation of integral dependence of $\overline{\varphi^*}(f)$ over $\mathcal{O}(Z)$

$$\overline{\varphi^*}(f)^n + b_1 \overline{\varphi^*}(f)^{n-1} + \cdots + b_n = 0, \quad b_i = \varphi^*(a_i) \in \mathcal{O}(Z),$$

and since $\mathcal{O}(Z)$ is integrally closed, we must have $\overline{\varphi^*}(f) \in \mathcal{O}(Z)$. Thus, we have $\alpha = \overline{\varphi^*}|_{\mathcal{O}(\widehat{Y})}$ as the desired map, and it is unique by construction. Hence, we have a unique morphism $\theta : Z \rightarrow \widehat{Y}$ such that $\varphi = \pi \circ \theta$ and $\theta^* = \alpha : \mathcal{O}(\widehat{Y}) \rightarrow \mathcal{O}(Z)$ by (3.5).

□

20. Let Y be a variety of dimension ≥ 2 , and let $P \in Y$ be a normal point. Let f be a regular function on $Y - P$.

- (a) Show that f extends to a regular function on Y .
- (b) Show this would be false for $\dim Y = 1$.

Proof.

- (a) Y can be covered by open affine varieties, so we reduce to the case when $Y \subseteq \mathbb{A}^n$ itself is an affine variety. Then by composing with the inclusion morphism $Y \rightarrow \mathbb{A}^n$, we further reduce to the case when $Y = \mathbb{A}^n$, that is we want to show if $f = g/h$ on $\mathbb{A}^n - P$, then we can extend f to a regular function on Y . We show that if a polynomial $h \in k[x_1, \dots, x_n]$ is nonzero on $\mathbb{A}^n - P$, then f is nonzero on P . We can also assume $P = (0, \dots, 0)$ and $n = 2$ by induction. Suppose $h(P) = 0$, then h must have zero constant term, so we can write

$$h(x, y) = f_0(x)y^d + f_1(x)y^{d-1} + \cdots + f_d(x), \quad f_i(x) \in k[x] - k,$$

then there exists nonzero $a \in k$ such that $f_i(a) \neq 0$ for some i , we have

$$h(a, y) = a_0 y^d + a_1 y^{d-1} + \cdots + a_d \neq 0, \quad a_i = f_i(a),$$

and since k is algebraically closed, $h(a, y)$ must have a solution $y = b$, so (a, b) is a solution to h that is not equal to P , a contradiction.

- (b) Consider $Y = \mathbb{A}^1$ and $P = 0$, and let $f = \frac{1}{x}$. Then f is regular on $Y - P$ by definition, but it clearly cannot be extended to the entire affine line.

□